

General Education Reflection

As I reminisce on my college career, I can look back on all of the engineering courses I've taken and can accurately point out how those classes made me a better engineer. However, while taking my general education classes, which are outside the scope of my engineering degree, I have had to think a little harder to connect what I was learning to my other courses and experience. In doing so I have concluded that just because a class isn't strictly about engineering or problem solving, good takeaways can still be had.

Having started Iowa State with nine general education credits already fulfilled, I only needed to take two more classes to finish the requirement. The first such course I took was named Russia Today. Included in the learning for this course was a survey of social, political, economic, and cultural topics relevant to contemporary Russia, from WW2 to the present day. I learned about the leaders of the U.S.S.R and their policies, the societal state of the country, and the geopolitical landscape of the time. Multiple case studies and live interviews with citizens of the Soviet Union shed light on the real day-to-day life of someone living in Russia at the time.

The second general education course I've taken was named The History of American Rock and Roll. In this course I learned about the development of rock styles from their roots in blues and folk music, and various overlaps and crossovers with other musical traditions including hip-hop, country, and pop. I also built listening experience through study of song forms, musical instruments of rock, and socio-political influences. I used these skills to recognize artists and genres simply by listening to a short clip of a song.

There are a couple ways I've applied the learning these classes have given me. One of which is the background on Russian or Russia-backed cybercriminals. Today, these individuals or groups play a prevalent role in the information security landscape, so knowing the world they came from is a good way to know how they think and what they value. One of the best ways to deter cyberattacks is to think like your aggressor, so Russia Today helped give me the context and history needed to think like a Russian and possibly stop a cyberattack.

I've also learned to apply skills I learned in History of American Rock and Roll. One way is that I needed to recognize patterns and structures in something as complex as a song. This is the same part of the brain used in reverse engineering, malware analysis, or network traffic analysis. Strengthening my pattern recognition is imperative to doing these things fast, efficiently and correctly. I also learned that technology, like engineering products, doesn't exist in a cultural vacuum. Issues such as data privacy, algorithmic bias, or the digital divide are all engineering problems that also carry real social and cultural weight. Understanding how music is shaped and was shaped by society mirrors how engineers must consider social context when building systems.

Over my career at Iowa State, I've had the opportunity to take a wide array of different courses. From Chemistry to Digital Forensics, all the classes I've been in have helped me expand my knowledge in a variety of different ways. Different from the others, however, were my general education courses. In the others, the benefit to my education was clear. They directly improved my

cybersecurity, intellect and knowledge. Taking the general education classes made me think about how I could apply what I was learning in a different way than the rest of my classes. This led me to be a better, more well-rounded student and person.